

Apolipoprotein *N*-acyltransferase (Lnt): the phospholipid-dependent final step of bacterial lipoprotein maturation

A review-style synthesis anchored on the Pseudomonas putida KT2440 UniPathway UPA00666 module (Lnt = PP_4790, UniProtKB:Q88DN4)

1. Executive summary

Bacterial lipoproteins are anchored in the cell envelope through a lipid-modified N-terminal cysteine and serve in envelope assembly, transport, enzymatic catalysis, signalling and virulence. Their maturation is a strictly ordered, three-enzyme, membrane-embedded post-translational pathway: **Lgt** → **LspA** → **Lnt**. **Apolipoprotein *N*-acyltransferase (Lnt)** catalyses the *final* step — transfer of a single acyl chain from a membrane glycerophospholipid to the free α -amino group of the already diacylated N-terminal cysteine — converting a *diacyl* apolipoprotein into the mature *triacyl* lipoprotein

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In diderm (Gram-negative/proteobacterial) cells this is a chemically and topologically well-defined reaction: Lnt is an integral inner-membrane enzyme of the **carbon-nitrogen (CN) hydrolase / nitrilase superfamily** that operates by a **ping-pong mechanism** through a covalent **thioester acyl-enzyme intermediate** on a catalytic cysteine, using an **E267-K335-C387 catalytic triad** (numbering from *E. coli*)

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Crystal structures of Lnt from *Pseudomonas aeruginosa* and *Escherichia coli* are near-identical — a multi-TM membrane domain supporting a periplasmic catalytic domain with an active site at the membrane-periplasm interface — and provide the best available structural proxy for

the *P. putida* KT2440 enzyme

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The system's boundaries are important. Lnt maturation is **upstream of, and mechanistically distinct from, the Lol trafficking pathway** (LolCDE–LolA–LolB) that sorts mature lipoproteins to the outer membrane; N-acylation is a *prerequisite* for efficient Lol recognition but is not itself a sorting step

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Across evolution, Lnt is conserved in proteobacteria and in high-GC Gram-positives (Actinobacteria/Mycobacteria) but is **absent from low-GC Firmicutes**, which nonetheless N-acylate their lipoproteins using non-orthologous, convergent enzymes and produce a wider range of acylation states

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Lnt is **essential in proteobacteria** and is an antibiotic-target candidate, though a more challenging one than LspA

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P. putida KT2440, a γ -proteobacterium, is expected to use the canonical, essential Lnt route.

2. Definition and biological boundaries

2.1 What the system is

The UPA00666 module is narrowly the **N-acylation reaction that completes lipoprotein maturation**:

- **Substrate (acceptor):** S-diacylglyceryl-cysteine apolipoprotein — a lipoprotein whose N-terminal cysteine already bears a thioether-linked diacylglyceryl group (installed by Lgt) and whose signal peptide has been removed (by LspA), leaving a free α -amino group.
- **Co-substrate (acyl donor):** a membrane **glycerophospholipid**, predominantly phosphatidylethanolamine (PE) in enterobacteria
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- **Enzyme:** Lnt (PP_4790 / Q88DN4 in *P. putida* KT2440), an integral inner-membrane protein.
- **Product:** the **mature triacylated lipoprotein**, carrying two ester-linked acyl chains (diacylglyceryl) plus one **amide (N)-linked** acyl chain on the cysteine α -amino group, and a by-product **lyso-phospholipid**.

The module therefore *begins* with the LspA product and *ends* with the triacyl lipoprotein. It is defined by a single covalent chemistry — amide bond formation on an N-terminal cysteine — carried out by one enzyme.

2.2 Neighbouring processes that should be treated separately

Several adjacent processes are frequently discussed together with Lnt but are mechanistically distinct and should not be conflated:

1. **Upstream maturation steps.** *Lgt* (phosphatidylglycerol:prolipoprotein diacylglyceryl transferase) recognises the **lipobox** and adds the diacylglyceryl group; *LspA* (signal peptidase II) removes the signal peptide

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These generate Lnt's substrate but are separate enzymes/reactions.

2. **Lol trafficking.** The **LolCDE** ABC transporter extracts OM-destined lipoproteins from the inner membrane, hands them to the periplasmic chaperone **LolA**, which delivers them to the OM receptor **LolB**

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This is a *transport* system, not a modification step; it acts on Lnt's product. Notably, LolB is restricted to γ -proteobacteria and absent from some diderms (e.g., spirochetes), underscoring that trafficking machinery varies independently of the acylation chemistry

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3. **Sorting rules (the "+2 rule")**. Whether a mature lipoprotein stays in the inner membrane or is exported depends on the residue immediately following the acylated cysteine (classically Asp at +2 acts as a Lol-avoidance/IM-retention signal in *E. coli*)

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This is a sequence-encoded sorting determinant, orthogonal to the acylation reaction.

4. **Innate-immune recognition**. Triacylated versus diacylated lipopeptides are differentially sensed by TLR2/TLR1 vs TLR2/TLR6. This is a host-side consequence of the acylation state, not part of the bacterial enzymatic module

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2.3 Competing definitions

There is broad agreement on the biochemistry in proteobacteria. The main definitional tension concerns **"apolipoprotein N-acyltransferase" as a function versus Lnt as a specific protein family**: low-GC Firmicutes perform N-acylation without an Lnt ortholog, so "apolipoprotein N-acyltransferase activity" is polyphyletic even though the *lnt* gene family is not universal

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For UPA00666 in *P. putida*, the narrow, protein-specific definition (Lnt = CN-hydrolase-family enzyme) applies.

3. Mechanistic overview

3.1 The best current model: a two-step ping-pong transacylation

Lnt runs a **bi-bi ping-pong** reaction in two half-reactions (P 21676878).

1. **Acyl-enzyme formation (donor half-reaction).** The catalytic cysteine (C387 in *E. coli*) attacks the *sn*-1 ester carbonyl of a glycerophospholipid, forming a **covalent thioester acyl-enzyme intermediate** and releasing a **lyso-phospholipid**. This half-reaction is comparatively *slow* and rate-influencing.
2. **N-acyl transfer (acceptor half-reaction).** The α -amino group of the diacylglyceryl-cysteine apolipoprotein attacks the thioester, resolving the intermediate and installing the **amide-linked** acyl chain. This step is *fast* in vitro.

The **E267-K335-C387** triad is the catalytic core: Cys is the nucleophile, and the Glu/Lys pair supports thioester formation and aminolysis, consistent with nitrilase/CN-hydrolase chemistry (P 17416655).

Structural snapshots subsequently visualised intermediate states along this pathway, reinforcing the interfacial, two-step model (P 37390210).

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3.2 Which steps are obligatory, conditional, or accessory

- **Obligatory:** (i) prior diacylglyceryl modification and signal-peptide cleavage (Lnt cannot act on an uncleaved prolipoprotein or a non-diacylated cysteine); (ii) acyl-enzyme formation *before* N-acyl transfer (enforced by the ping-pong mechanism); (iii) an intact catalytic triad and the additional essential residues Y388/E389 (active-site pocket) and W237/E343 (mobile arms) (P 17416655).
- **Conditional / substrate-dependent:** the *identity* of the transferred acyl chain, which reflects the cellular phospholipid pool and headgroup/acyl-chain composition rather than a

fixed intrinsic selectivity

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- **Accessory / downstream:** Lol-mediated sorting and the +2 retention rule — required for correct localisation of the product but not for the acylation chemistry itself

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3.3 Topology and compartmentalisation

Lnt is a polytopic inner-membrane protein; *E. coli* topology mapping indicates **six transmembrane segments** with the catalytic domain facing the **periplasm**

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Structures show the periplasmic globular (CN-hydrolase) domain sitting atop the membrane domain, with the active site at the interface — ideally placed to accept a membrane-embedded phospholipid donor and a membrane-anchored lipoprotein acceptor while performing chemistry in the aqueous periplasmic environment

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4. Major molecular players and active assemblies

Step	Enzyme / assembly	Reaction	Notes
1	Lgt	Diacylglyceryl transfer to lipobox Cys (from PG)	Recognises lipobox; generates thioether-linked diacylglyceryl-cysteine (P 28698362)
2	LspA (signal peptidase II)	Signal-peptide cleavage	Yields free N-terminal S-diacylglyceryl-Cys; top antibiotic target (globomycin, myxovirescin) (P 31036406)
3	Lnt (this module; PP_4790/Q88DN4)	N-acylation of Cys α -amino group from a phospholipid	CN-hydrolase family; E267-K335-C387; thioester intermediate; ping-pong (P 21676878) (P 17416655)
Downstream	LolCDE / LolA / LolB	IM→OM sorting of mature lipoprotein	Separate transport module; depends on Lnt product; LolB γ -proteobacteria-restricted (P 15513925) (P 39149330)

Structural anchor for *P. putida*: The *P. aeruginosa* and *E. coli* Lnt crystal structures are "remarkably similar"

([P 28675161](#)).

Because *P. putida* KT2440 Lnt is a close γ -proteobacterial relative of *P. aeruginosa* Lnt, and proteobacterial Lnt orthologs cross-complement in vivo

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the *P. aeruginosa* structure is the best representative for understanding PP_4790's architecture and active site. No experimental *P. putida* Lnt structure is required to infer the mechanism, though this remains an inference rather than a direct measurement.

Acyl donor: glycerophospholipids, with PE the efficient donor in enterobacteria and activity strongly modulated by headgroup and acyl-chain composition (P 21676878).

P. putida membranes are PE/PG-rich, so PE is the plausible physiological donor, but this has not been directly demonstrated in *P. putida* and is extrapolated from *E. coli*.

5. Evolutionary and cell-biological variation

5.1 Deep origin and family membership

Lnt belongs to the **nitrilase / carbon-nitrogen hydrolase superfamily**, an ancient and widespread group of Cys-Glu-Lys catalytic-triad enzymes (P 17416655).

(P 29859843). The lipoprotein-processing pathway as a whole is a conserved feature of the bacterial envelope, and the diacyl→triacyl chemistry appears deeply rooted in diderm lineages. Within the superfamily, the **soluble nitrilases are the best out-group for understanding the ancestral catalytic core**, while the membrane domain and mobile "arms" are Lnt-specific elaborations that adapt the enzyme to interfacial, lipid substrates (P 29859843).

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5.2 Lineage variation and alternative routes

- **Proteobacteria (incl. *P. putida*):** canonical, essential Lnt; triacylated lipoproteins; product feeds the Lol pathway

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- **High-GC Gram-positives (Actinobacteria/Mycobacteria):** possess functional Lnt (e.g., *M. bovis* BCG_2070c); knockout abolishes N-acylation; lipoproteins are triacylated, sometimes with tuberculostearic/palmitoyl chains

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These Lnt do **not** cross-complement *E. coli*, indicating divergent substrate/partner surfaces

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- **Low-GC Gram-positives (Firmicutes):** no Lnt ortholog, yet N-acylation occurs via **non-orthologous, convergent** enzymes, and the N-terminal acylation state is **structurally heterogeneous** (di-, tri-, N-acetyl, lyso forms) — a genuinely alternative molecular route to a similar chemical outcome

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In some species N-acylation depends on membrane lipid state/physiology (e.g., pH/lipid composition), a regulatory dimension largely absent from the essential proteobacterial route

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5.3 Physiological/compartamental variation

Because Lnt acts at the inner-membrane/periplasm interface, its activity is embedded in envelope lipid homeostasis: the acyl chains it transfers mirror the phospholipid pool

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Variation is therefore expected across growth conditions that reshape membrane lipids, but for the essential proteobacterial enzyme the reaction chemistry itself is invariant.

6. Constraints, dependencies, and failure modes

- **Strict pathway order.** Lnt requires the *product of LspA* (a cleaved, diacylated cysteine with a free α -amino group); it cannot act on a prolipoprotein or an unmodified cysteine. The three enzymes act sequentially and non-interchangeably

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- **Internal reaction order.** The ping-pong mechanism mandates acyl-enzyme formation *before* aminolysis; the two half-reactions cannot be reversed

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- **Compartmental constraint.** Catalysis occurs at the periplasm-facing membrane interface via a six-TM enzyme; both substrates are membrane-associated while the chemistry is periplasmic

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- **Downstream dependency.** N-acylation is required for efficient **Lol recognition**; loss of Lnt causes apo-lipoproteins (e.g., apoLpp) to accumulate in the inner membrane and is **lethal** in *E. coli* — evidence that ruling-out an "unacylated export" path is well supported

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- **Failure modes.** Depletion/inactivation of Lnt → accumulation of diacyl apolipoproteins, mislocalisation, envelope stress, and death in proteobacteria

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This essentiality is the basis for antibiotic interest, tempered by the difficulty of achieving selectivity at Lnt's active site

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7. Controversies and open questions

1. **Phospholipid donor selectivity.** Early work suggested minimal headgroup preference; later defined in vitro assays found strong headgroup/acyl-chain dependence (PE efficient) (P 21676878).
The physiological donor(s) in *P. putida* specifically have not been measured and are inferred from enterobacterial data.
2. **Direct evidence in *P. putida*.** The KT2440 Lnt (PP_4790) is assigned by orthology/UniPathway; mechanism, structure, essentiality, and donor identity are extrapolated from *E. coli* and *P. aeruginosa*. Direct biochemical/genetic validation in KT2440 is lacking. This is a clear over-extrapolation risk the field should acknowledge.
3. **Structural interpretation of intermediates.** Multiple 2017 structures and later "snapshot" work support the interfacial ping-pong model, but details of substrate entry/exit routes and arm dynamics (W237/E343) remain partly modelled rather than fully resolved (P 28675161 P 28698362 P 37390210 P 17416655).
4. **Cross-lineage comparability.** Data are mixed across *E. coli*, *P. aeruginosa*, mycobacteria, and Firmicutes. Because actinomycete Lnt does not complement *E. coli* and Firmicutes lack Lnt entirely, mechanistic claims should not be generalised uncritically across all bacteria (P 17416655 P 24093492 P 32723923).
5. **Druggability.** Whether Lnt (as opposed to LspA) can yield a selective, non-toxic antibacterial remains open; its buried, membrane-interfacial active site is a practical obstacle

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6. **Regulation.** Whether Lnt activity is regulated (versus constitutively essential) — as hinted by lipid/physiology-dependent N-acylation in some Gram-positives — is unresolved for proteobacteria

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Scope note: This review treats the **proteobacterial, Lnt-dependent** N-acylation module (UPA00666) as the core system, with *P. putida* KT2440 Lnt (PP_4790/Q88DN4) as the focal ortholog. Claims specific to KT2440 are, at present, inferences from close relatives (*P. aeruginosa*, *E. coli*) rather than direct measurements, and are flagged as such throughout.
